

Matias Freire

551-331-8011 | matiaspfreire@gmail.com | [Github](#) | [Linkedin](#) | [Portfolio-Website](#) | NY Metropolitan Area

EDUCATION

Stevens Institute Of Technology

Expected May 2028

B.S. in Computer Science

Hoboken, NJ

Coursework: Algorithms and Data Structures, Computer Architecture, Systems Programming, Software Development Processes

Certifications: Bloomberg Market Concepts, Hugging Face AI Agents Fundamentals

Cornell Tech – Break Through Tech

Summer 2026

Machine Learning Fellow

Remote

- Building advanced skills in machine learning, deep learning, and neural networks through an intensive Cornell Tech program, culminating in the AI Studio Fellowship , a paid team-based placement partnering with industry sponsors to develop real-world ML solutions.

EXPERIENCE

Software Engineering Intern - MCA Team

May 2026 – Present

Wabtec Corporation

Erie, PA

- Developing a intelligent panel to visualize real-time train data by integrating Data Distribution Service (DDS) communication with embedded system workflows within an WSL Enviroment development.
- Encapsulating locomotive data monitoring functionality into a user-facing application by building a custom UI with Python RTI API, Tailwind CSS, HTML and Docker.

Deep Learning Research Assistant

Aug 2025 – Mar 2026

Advanced Quantum Materials Lab

Hoboken, NJ

- Developed a Dash-Plotly platform integrating literature-driven preprocessing, synthetic data, and open-source pipelines for interactive MoS defect visualization and DL training.
- Studied preprocessing methods, synthetic data generation, and deep learning models (CNNs like MobileNet), Extraordinary Magnetoresistance (EMR) devices.

PROJECTS

SHPE and Colorstack Stevens Chapter Website | *Next.js, React, Tailwind, Typescript*, Supabase [Github-Repo](#)

- Built from scratch modular, responsive websites for the SHPE and ColorStack Stevens chapters, **serving 50+ and 150+ members as central platforms** to attract corporate sponsors and onboard new members
- Implemented reusable UI components in TypeScript and Tailwind CSS, integrating **authentication and CRUD** functionality, and **leveraged the Supabase MCP within Google's Antigravity environment** to accelerate full-stack development

MoS2 Image Synthesis & Analysis Platform | *PyTorch, Dash-Plotly, CSS, Docker*

[Github-Repo](#)

- Designed and built from scratch an end-to-end Dash-Plotly research platform for deep learning-based MoS defect analysis
- Automated synthetic image generation with configurable defects and realistic noise injection, integrating a ResUNet model for vacancy and polymorph classification

LEADERSHIP

Vice-President - Previous Head of External Relations | *Colorstack Stevens Chapter*

April 2025 – Present

- Led growth and technical engagement initiatives, **increasing membership by 150% and online engagement by 125% through social media** strategy and **15+ events per semester**
- Serve as the primary representative for partnerships and collaborations with major campus organizations, including CS Club, SHPE, NSBE, and others
- Organized peer-led weekly technical sessions, guiding members through 32+ Leetcode Questions

Chief Web Officer | *SHPE Stevens Chapter*

March 2026 – Present

- Leading development of the official chapter website in Next.js/Supabase, building the platform from scratch to **support 150+ registered members** and present the organization to corporate sponsors
- lanning and organizing technology-focused events and workshops to expand technical engagement across the chapter

STEP Tutor-Counselor/Resident Assistant | *STEP Bridge Summer Program*

Jun 2025 – Aug 2025

- Led Calculus I/II and Physics recitations for **52+ incoming freshmen**, strengthening STEM readiness through structured reviews and weekly after-hours **tutoring for 25+ student**
- Served as a **Resident Assistant, mentoring 52+ residents** while enforcing residential policies and supporting academic success, wellness, and community development

TECHNICAL SKILLS

Programming: Java, Python, C/C++, JavaScript

Web Tools: TypeScript, React, Tailwind, Next.js, Supabase

Libraries: PyTorch, Scikit-learn, Dash-Ploty, RTI Python Connex, Pandas, NumPy

Developer Tools: Git, GitHub/Gitlab, Docker , Jupyter Notebooks, Linux, RTI Panel